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FedDHKD: A Lightweight Federated Learning Framework Based on Dynamic Hierarchical Knowledge Distillation for 5G IoT

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***Abstract*—With the rapid proliferation of 5G technology and the exponential growth of IoT devices, federated learning is emerging as a typical paradigm in 5G IoT scenarios. However, traditional federated learning still faces numerous challenges in 5G IoT scenarios, including resource constraints on edge devices, strong data distribution heterogeneity, and stringent privacy protection requirements. To address these challenges, this paper proposes a lightweight federated learning privacy protection framework (FedDHKD) based on dynamic hierarchical knowledge distillation. The FedDHKD framework divides the model into a client-side feature extractor and a server-side classifier, and achieves efficient knowledge transfer and model compression through a dynamic knowledge distillation mechanism, thereby reducing communication overhead. Additionally, an optimization strategy based on model-agnostic meta-learning is introduced to enhance adaptability to heterogeneous data distributions and accelerate convergence. Furthermore, by adding Gaussian noise that satisfies differential privacy constraints to the low-dimensional features to be uploaded, lightweight and effective privacy protection is achieved. Finally, we evaluated the performance of the FedDHKD framework against baseline approaches across various datasets. Experimental results demonstrate that FedDHKD reduces communication overhead to approximately one-seventh of FedAvg while maintaining competitive model accuracy. This approach provides an effective solution for 5G IoT edge scenarios, achieving a balanced trade-off between performance, efficiency, and privacy.**

***Index Terms*—5G Internet of Things, Federated learning, Knowledge distillation, Lightweight model, Differential privacy.**

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I. Introduction

The development of edge intelligence and distributed learning is being propelled by the exponential growth in the number of IoT devices brought about by the quick rollout of 5G technology [1] [2]. With its wide coverage, low latency, and high bandwidth, 5G networks offer IoT devices strong communication support, facilitating the effective transmission and processing of enormous volumes of data at the edge. Federated learning enables multiple parties to collaborate on training without exchanging raw data. By distributing model training tasks to local devices and aggregating model parameters to a central server, federated learning reduces data transmission pressure and opens up new avenues for user privacy protection [3]. Figure 1 shows how federated learning is gradually becoming an important solution in 5G IoT scenarios.

Although federated learning shows great potential in 5G IoT scenarios, its practical application still faces many challenges [4]. Due to the limitations of computing power, storage capacity, and communication bandwidth of IoT devices, traditional complex privacy protection technologies are difficult to directly apply to these resource-constrained edge devices. The strong heterogeneity of data distribution is another critical issue. In IoT environments, data generated by different devices often exhibit significant differences, and this non-independent and identically distributed (Non-IID) characteristic may lead to degraded model training performance or even convergence difficulties [5]. Additionally, it is challenging to address a variety of needs with a single model architecture since edge device hardware

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configurations and task requirements vary widely. Federated learning is made much more challenging by stringent privacy protection. User data security is crucial in 5G IoT situations, but current federated learning techniques are still ineffective against threats like man-in-the-middle and gradient leakage attacks [6]. These problems show that the requirements of 5G IoT scenarios cannot be addressed by existing federated learning techniques, and a lightweight and effective privacy protection framework is desperately needed to overcome these obstacles [7].

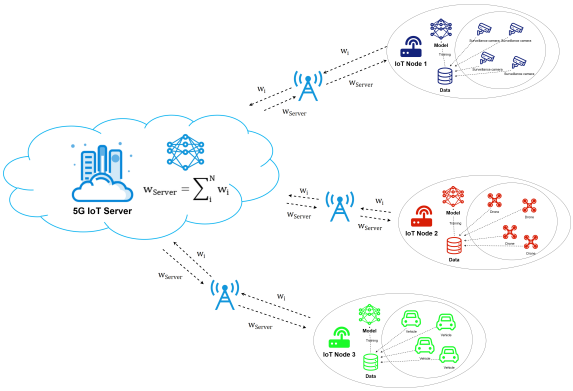


Fig. 1. Schematic diagram of a federated learning system in a 5G IoT scenario.

Addressing the challenges faced by federated learning in 5G IoT scenarios, this paper proposes a lightweight federated learning privacy protection framework based on dynamic hierarchical knowledge distillation, combining model partitioning, dynamic hierarchical knowledge distillation, heuristic optimization strategies inspired by model-agnostic meta-learning, and differential privacy techniques. The framework achieves functional separation and lightweight training of models through model partitioning and dynamic knowledge distillation; it introduces an optimization strategy based on model-agnostic meta-learning to enhance the model's adaptability to new data distributions and accelerate convergence; additionally, the framework injects differential privacy noise into the low-dimensional feature space, thereby improving privacy protection performance while maintaining model performance. In summary, the main contributions of this paper are as follows:

- This paper proposes a lightweight federated dynamic hierarchical knowledge distillation framework, FedDHKD, which decouples the functions of the client and server by dividing the model into a low-level feature extraction module and a high-level classification module. The client only deploys a lightweight feature extraction model, which is responsible for extracting and uploading local data features. The server maintains a high-performance high-level classification model, which generates soft logits based on aggregated features and distributes them to guide the knowledge distillation process on the client side. This framework eliminates the need for clients to pre-train and deploy local teacher models, as all supervision signals required for distillation are uniformly provided by the server, significantly reducing the computational and deployment complexity on the client side. Additionally, the framework introduces an adaptive adjustment mechanism for distillation temperature to dynamically control the strength of knowledge transfer, further enhancing model training efficiency and generalization capabilities.
- FedDHKD introduces an optimization strategy based on Model-Agnostic Meta-Learning (MAML) to enhance the global model's adaptability to the local data distribution of client-side devices. The server simulates the client's optimization process to construct a meta-training task, training a global teacher model with "prior adaptability." This strategy effectively mitigates the common model drift issue in traditional federated learning methods and exhibits good model structure agnosticism, making it suitable for various network architectures and task scenarios.
- This paper integrates a feature-level differential privacy mechanism into FedDHKD to enhance privacy protection during data uploading. By injecting Gaussian noise that satisfies the (ϵ, δ)-differential privacy constraint into low-dimensional features, potential data leakage risks are effectively prevented. In addition, the residual structure design improves the model's robustness to noise, achieving a good balance between privacy protection and model performance.

The structure of this paper is as follows: Section 2 introduces related research work. Section 3 defines the problem description. Section 4 introduces the FedDHKD framework and provides detailed explanations of its specific design details. Section 5 describes the experimental setup and discusses the experimental results. Section 6 summarizes this paper.

II. Related Works

A. Federated Learning and Privacy Protection Technology

Federated learning is a distributed machine learning framework that enables cross-institutional collaborative modeling by allowing participants to train models locally and share only model parameters. Its core advantage lies in addressing data silo issues while complying with privacy protection regulations. Kim et al. [8] proposed a subgraph federated learning framework, which achieves localized generalization by decomposing graph data substructures, thereby reducing the global model's dependence on individual client data.The FedDM framework proposed by Sun et

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al. [9] optimizes communication efficiency through iterative distributed matching technology, reducing communication rounds by 90% in image classification tasks while maintaining model accuracy.

However, the distributed nature of federated learning also introduces new security challenges, especially the risk of gradient privacy leakage. In federated learning, clients need to upload model gradients to the server to complete the aggregation update. Although gradients appear to be intermediate products of model training, research shows that they may leak original training data. Therefore, it is necessary to combine some privacy protection technologies to provide more reliable privacy protection.Shukla et al. [10] jointly optimized federated learning and differential privacy in a medical setting. By injecting Gaussian noise (privacy budget ϵ=1.9) into model updates, they achieved a balance between privacy and accuracy in breast cancer diagnosis tasks. To address data heterogeneity issues, they proposed a weighted aggregation strategy to ensure that the model supports multi-institutional collaboration in compliance with HIPAA/GDPR requirements without sharing raw data.The RoFL framework proposed by Lycklama et al. [11] significantly improves the robustness of federated learning in a single-server setting by introducing normative constraints and a lightweight packet loss resilience protocol, while supporting secure aggregation.The two-tier data packing technology proposed by Zhou et al. [12] significantly reduces the computational and communication overhead in federated learning by optimizing the encoding method of RLWE homomorphic encryption.

B. Lightweight federated learning in 5G IoT scenarios

In the field of machine learning, model lightweighting has become a core requirement for 5G IoT scenarios, with core technologies including quantization, pruning, and distillation. Quantization significantly reduces model size and computational complexity by lowering the numerical precision of model parameters [13]. Pruning achieves model compression by removing redundant weights or neurons from neural networks [14]. Distillation transfers knowledge from large teacher models to lightweight student models, reducing inference overhead while maintaining performance [15] [16]. These methods are particularly important in resource-constrained 5G IoT edge devices. Butt et al. [17] explored the evolution direction of 5G-Advanced in 3GPP Release 19, including the introduction of lightweight technologies such as Ambient IoT. Ambient IoT is regarded as the missing link in the 3GPP IoT device domain, and it is expected to drive the popularization and large-scale application of IoT devices through its lightweight design and low-power characteristics.

Traditional federated learning also faces challenges in 5G IoT environments due to heterogeneous data distribution and differences in device computing power. The integration of lightweight technologies with federated learning is necessary for two reasons: feasibility under computing constraints and strict privacy protection requirements. On the one hand, 5G IoT edge devices have limited computing resources, and lightweight technologies enable federated learning to run efficiently on low-power devices. On the other hand, the combination of federated learning and lightweight technologies will bring efficient privacy protection.He et al. [18] proposed the FedGKT method, which periodically transfers knowledge from a small CNN to a large server-side CNN through knowledge distillation, enabling the training of large CNNs on edge devices. This method reduces communication costs and lowers the demand for edge computing.Caldas et al. [19] proposed methods using lossy compression techniques to reduce model size and communication costs, as well as Federated Dropout techniques, which allow users to perform effective local training on smaller subsets of the global model. These methods reduce the resource requirements of federated learning on client devices.Li et al. [20] proposed a lightweight federated learning method for NILM on end devices by combining neural architecture search (NAS) and federated learning. First, they studied a memory-efficient NAS method to determine personalized models under the resource constraints of end devices. Second, they designed a federated mutual learning method to coordinate the collaboration between distributed end devices and heterogeneous personalized models in a privacy-preserving manner.

III. Problem Formulation

This paper investigates a federated learning system comprising one IoT server and N IoT smart devices, where each IoT smart device $i(1 <= i <= N)$ possesses a local private dataset $\mathcal{D}_i$. The system aims to efficiently train a global shared machine learning model through collaborative training, thereby achieving efficient model training while protecting user privacy. Each IoT smart device typically has computational capabilities, limited communication bandwidth, and local data privacy protection requirements, making it suitable for intelligent decision-making and perception tasks in resource-constrained scenarios.

The training process of federated learning consists of multiple communication rounds. In each round, the server selects K IoT smart devices from N IoT smart devices to participate in training (usually $K \ll N$) and distributes the current global model $\mathbf{w}_t$ to the selected IoT smart devices. Subsequently, each IoT smart device trains the model locally based on its local dataset $\mathcal{D}_i$, aiming to minimize its local empirical risk function:

$$\min_{\mathbf{w}} \left[f_i(\mathbf{w}) = \frac{1}{|\mathcal{D}_i|} \sum_{(\mathbf{x},y)\in\mathcal{D}_i} \ell(\mathbf{w};\mathbf{x},y) \right] \tag{1}$$

where $\ell(\cdot)$ is the loss function (e.g., cross-entropy loss).

After training is complete, the IoT smart device uploads the updated model parameters $\mathbf{w}_{i,t+1}$ to the IoT server,

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which aggregates the models using a weighted average based on the proportion of data from each IoT smart device:

$$\mathbf{w}_{t+1} = \sum_{i=1}^{K} \frac{|\mathcal{D}_i|}{\sum_{j=1}^{K} |\mathcal{D}_j|} \mathbf{w}_{i,t+1} \tag{2}$$

Subsequently, the updated global model $\mathbf{w}_{t+1}$ is distributed to the next batch of selected IoT smart devices, and the above process is repeated until the predefined number of training rounds is reached or the model converges.

In a 5G IoT environment, the design of federated learning systems needs to balance the following three key objectives:

- **Maximizing model performance:** The central goal of federated learning is to obtain a globally shared model with strong generalization performance. This is formulated as the minimization of the global objective function:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{i=1}^{N} \frac{|\mathcal{D}_i|}{D} f_i(\mathbf{w}) \tag{3}$$

  where $D = \sum_{i=1}^{N} |\mathcal{D}_i|$ is the total data volume of all IoT smart devices in the system.
- **Minimizing communication overhead:** Due to the bandwidth constraints and energy limitations of edge IoT devices, communication efficiency becomes a critical bottleneck in federated learning systems. One of the key optimization objectives is to reduce the amount of data exchanged between clients and the server in each communication round. This can be formally expressed as:

$$\min \sum_{t=1}^{T} \sum_{i \in \mathcal{K}_t} \text{Comm}(\mathbf{w}_{i,t}) \tag{4}$$

  where $\mathcal{K}_t$ denotes the set of IoT smart devices participating in training in the tth round, and $\text{Comm}(\cdot)$ denotes the data volume of model parameters (e.g., in MB).
- **Privacy and security protection:** In 5G IoT scenarios, local data on clients often contain sensitive user information. Therefore, the system must incorporate privacy-preserving mechanisms such as differential privacy or secure aggregation to prevent potential leakage of private data through model updates, thereby ensuring end-to-end data privacy and system security.

IV. Methodology

In this section, we will provide a detailed technical overview of the FedDHKD framework. First, we designed a dynamic hierarchical knowledge distillation mechanism to compress the model, reduce communication overhead, and enhance privacy protection performance to a certain extent. Second, based on the concept of model meta-learning, we designed an optimization strategy to enhance the model's adaptability to client data distributions, significantly alleviating the model drift issues caused by Non-IID data distributions while improving training efficiency. Finally, by adding noise that satisfies differential privacy constraints to the low-dimensional features that need to be uploaded, we ensure data privacy and security.

A. FedDHKD Framework Design

To better adapt to the lightweight characteristics of edge intelligent devices in 5G IoT, the FedDHKD framework combines model partitioning, dynamic hierarchical knowledge distillation, model-agnostic meta-learning, and differential privacy technologies. Based on the above threat model, it redesigns the basic process of traditional federated learning with the following specific optimizations:

- Considering the limited computing, storage, and communication resources of edge intelligent devices, the FedDHKD framework achieves functional separation of the complete model through model partitioning, thereby deploying a lightweight low-level feature extraction model on resource-constrained clients and a more complex high-level classification model on resource-rich servers. Through model layering, the framework significantly reduces the overall overhead of clients to adapt to their limited resources.
- To better advance federated training under a model hierarchical architecture, the FedDHKD framework has designed a clever dynamic hierarchical knowledge distillation mechanism, treating the server-side high-level model as the teacher model and the client-side low-level model as the student model. Through the hierarchical knowledge distillation mechanism, the student model learns the output distribution of the teacher model, thereby achieving model compression and reducing communication overhead. Additionally, the framework designs a dynamic optimization algorithm that dynamically adjusts the distillation temperature and other hyperparameters to control the strength of knowledge transfer and improve training performance.
- Considering the issues of training efficiency and model drift caused by non-independent and identically distributed data, the FedDHKD framework designs optimization strategies based on model-agnostic meta-learning principles. The server constructs meta-training tasks by simulating the optimization process of clients, training a global teacher model with "prior adaptability" to enhance the global model's adaptability to the local data distribution of client devices and accelerate model convergence.
- To further enhance privacy protection performance, the FedDHKD framework adopts parameter-level differential privacy technology, which injects differential privacy noise into the low-dimensional feature space

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of the intermediate results of the client's low-level model. This satisfies the strict constraints of (ϵ, δ)-DP while avoiding performance degradation caused by parameter-level noise addition. The residual structure of the feature extractor retains shallow features through an identity mapping, enabling the model to remain stable during training even under noise interference.

B. FedDHKD Framework Design

The overall steps of the FedDHKD framework can be summarized as follows, as shown in Figure 2:

The pseudocode for the specific process details of the FedDHKD framework is shown below.

Algorithm 1: FedDHKD (Federated framework)

Input: $\{f_\theta(\cdot), g_\phi(\cdot), x_1, x_2, ..., x_i, \epsilon, \delta, \eta, \beta, \alpha, T\}$.
Output: Updated local models $\{f_{\theta_i}, g_{\phi_i}\}$ for each client i.

for round = 1 **to** MAXROUND **do**
  // IoT smart devices
  for each client $i = 1$ **to** N **do**
    Train low-level model $f_\theta(x)$ on local data $x_i \in \mathcal{D}_i$.
    Compute sensitivity Δf, and add Gaussian noise:
    $z_{dp}^{(i)} = f_\theta(x_i) + \mathcal{N}(0, \sigma^2)$, where $\sigma = \frac{\Delta f}{\epsilon} \cdot \sqrt{2 \ln\left(\frac{1.25}{\delta}\right)}$.
    Upload $z_{dp}^{(i)}$ to the IoT server.
  end for
  // IoT server
  Collect all noisy features and labels:
  $\mathcal{D} = \{(z_{dp}^{(i)}, y^{(i)})\}_{i=1}^{N}$.
  Construct meta-training tasks:
  $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_N\}$, where $\mathcal{T}_i = \{(z_{dp}^{(i)}, y^{(i)})\}$.
  for each task $\mathcal{T}_i$ **do**
    Perform local gradient update (inner update):
    $\phi_i' = \phi - \eta \cdot \nabla_\phi \mathcal{L}_{CE}(g_\phi(z_{dp}^{(i)}), y^{(i)})$.
    Compute meta-loss: $\mathcal{L}_{meta}^{(i)} = \mathcal{L}_{CE}(g_{\phi_i'}(z_{dp}^{(i)}), y^{(i)})$.
  end for
  Update global model using meta-loss:
  $\phi \leftarrow \phi - \beta \cdot \nabla_\phi \left(\frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{meta}^{(i)}\right)$.
  for each client $i = 1$ **to** N **do**
    Generate soft logits via server inference:
    $\hat{y}_{logits}^{(i)} = g_\phi(z_{dp}^{(i)})$.
    Transmit $\hat{y}_{logits}^{(i)}$ back to client i.
  end for
  // IoT smart devices
  for each client $i = 1$ **to** N **do**
    Receive $\hat{y}_{logits}^{(i)}$ from server.
    for each $T_m \in T_{\text{candidates}}$ **do**
      Simulate distillation based on temperature T_m:
      Train low-level model $f_\theta(x)$ via knowledge distillation:
      $\min_\theta \mathcal{L}_{KD} = \alpha \cdot \mathcal{L}_{distill} + (1 - \alpha) \cdot \mathcal{L}_{CE}$.
      Evaluate model performance $acc_{f_\theta(\cdot)}$.
    end for
    Select the optimal distillation temperature T_{best}.
    Update candidate pool dynamically.
    Train low-level model using selected T_{best}:
    $\min_\theta \mathcal{L}_{KD} = \alpha \cdot \mathcal{L}_{distill} + (1 - \alpha) \cdot \mathcal{L}_{CE}$.
  end for
end for
return $\{f_{\theta_i}, g_{\phi_i}\}_{i=1}^{N}$

Step 1: Model partitioning and initialization deployment: The neural network is divided into two submodels: a low-level feature extraction model $f_\theta(\cdot)$ and a high-level classification model $g_\phi(\cdot)$. The low-level feature extraction model $f_\theta(\cdot)$ is deployed on IoT smart devices participating in federated learning to extract high-dimensional feature representations $z = f_\theta(x)$ from the initial data x. The high-level classification model $g_\phi(\cdot)$ is deployed on the IoT server side, receiving the low-level features z uploaded from the IoT smart devices and outputting the final classification results $y = g_\phi(z)$.

Step 2: Local model training for IoT smart devices: Each IoT smart device i trains its local low-level feature extraction model $f_\theta(\cdot)$ based on private data x_i in its local dataset $\mathcal{D}_i$.

Step 3: Differential privacy feature noise: Before uploading low-dimensional features, each IoT smart device i adds Gaussian noise that satisfies the (ϵ, δ) differential privacy constraint, as shown in the following formula:

$$\begin{aligned} &z_{dp} = f_\theta(x) + \mathcal{N}(0, \sigma^2), \\ &\text{where} \quad \sigma = \frac{\Delta f}{\epsilon} \cdot \sqrt{2 \ln\left(\frac{1.25}{\delta}\right)} \end{aligned} \tag{5}$$

Step 4: IoT Server Aggregation and Meta-Learning Optimization: The IoT server aggregates all noisy features $z_{dp}^{(i)} i = 1^N$ uploaded by IoT smart devices and their corresponding labels $y^{(i)} i = 1^N$ to construct a global feature set $Z_{global} = z_{dp}^{(i)}, y^{(i)} i = 1^N$, and trains a global high-level model $g_\phi(\cdot)$ based on this, as shown below:

$$\min_\phi \mathcal{L}_{global} = \frac{1}{|Z_{global}|} \sum_{(z_{dp}, y) \in Z_{global}} \ell_{CE}(g_\phi(z_{dp}), y) \tag{6}$$

Specifically, it can be summarized into the following five parts:

- **The IoT server receives locally uploaded noise features and tags.**

$$\begin{aligned} &\mathcal{D} = \{(z_{dp}^{(i)}, y^{(i)})\}_{i=1}^{N} \\ &\text{where} \quad z_{dp}^{(i)} = f_\theta(x_i) + \mathcal{N}(0, \sigma^2) \end{aligned} \tag{7}$$

- **Constructing a set of meta-training tasks.** Divide dataset $\mathcal{D}$ according to the IoT smart devices participating in federated learning, and construct a set of meta-training tasks:

$$\begin{aligned} &\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_N\}, \\ &\text{where} \quad \mathcal{T}_i = \{(z_{dp}^{(i)}, y^{(i)})\} \end{aligned} \tag{8}$$

- **Perform a local gradient update (simulated local training) for each task.** For each task $\mathcal{T}_i$, calculate the local update parameters:

$$\phi_i' = \phi - \eta \cdot \nabla_\phi \mathcal{L}_{CE}\left(g_\phi(z_{dp}^{(i)}), y^{(i)}\right) \tag{9}$$

  Where η is the local learning rate.
- **Calculate the meta-loss of the updated model on the corresponding task.**

$$\mathcal{L}_{meta}^{(i)} = \mathcal{L}_{CE}\left(g_{\phi_i'}(z_{dp}^{(i)}), y^{(i)}\right) \tag{10}$$

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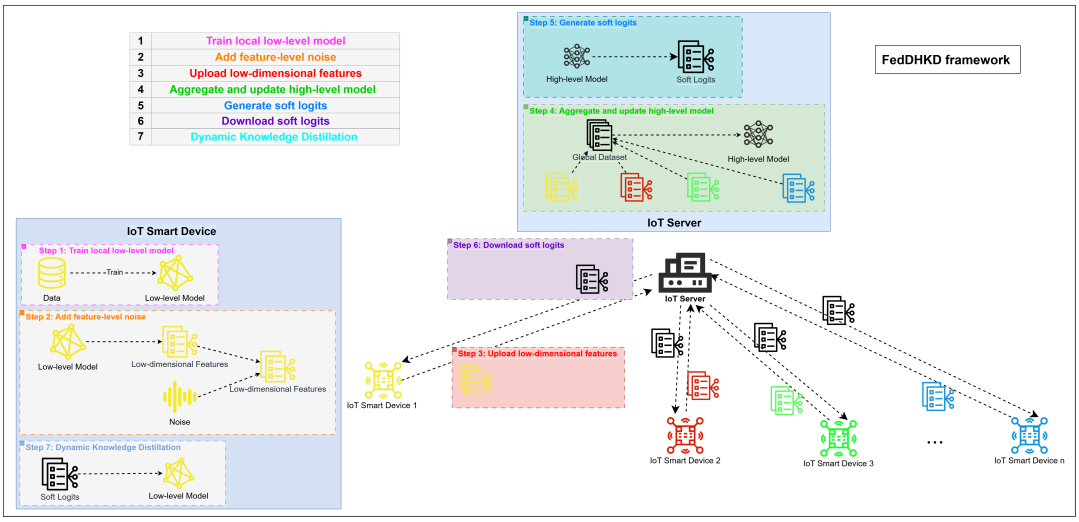


Fig. 2. Schematic Diagram of the FedDHKD Framework.

- **Update global model parameters.** Calculate the average meta loss for all tasks and update the global model:

$$\phi \leftarrow \phi - \beta \cdot \nabla_{\phi} \left(\frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{meta}^{(i)} \right) \tag{11}$$

Where β is the IoT server-side learning rate.

To address MAML's scalability issues, the IoT server constructs meta-training tasks using only 20% of IoT smart device features to reduce data processing volume. Furthermore, during the meta-learning process, only the top two layers of the IoT server's multilayer perceptron are updated.

Step 5: Lightweight knowledge distillation supervision signal delivery: The IoT server performs forward inference on the differentially private low-dimensional features z_{dp} uploaded by each IoT smart device and generates soft logits $\hat{y}_{logits} = g_{\phi}(z_{dp})$. These soft logits are then transmitted back to the respective IoT smart device as supervision signals for knowledge distillation.

Step 6: Lightweight dynamic hierarchical knowledge distillation: Each IoT smart device i directly uses the soft logits $\hat{y}_{logits}$ received from the IoT server to train its local low-level feature extraction model $f_{\theta}(\cdot)$. The distillation loss is computed between the server-generated soft distribution and the local model's output, allowing the IoT smart device to benefit from the knowledge of the global model without the need to maintain a teacher model locally.

The overall loss function remains a weighted combination of distillation and classification losses:

$$\min_{\theta} \mathcal{L}_{KD} = \alpha \cdot \mathcal{L}_{distill} + (1 - \alpha) \cdot \mathcal{L}_{CE} \tag{12}$$

Among them, $\mathcal{L}_{distill}$ is the distillation loss, and $\mathcal{L}_{CE}$ is the classification loss.

$$\mathcal{L}_{distill} = \frac{1}{|\mathcal{D}_i|} \sum_{(x,y)\in\mathcal{D}_i} \mathrm{KL} \left(\mathrm{Softmax}\left(\frac{\hat{y}_{logits}}{T} \right) \middle\| \mathrm{Softmax}\left(\frac{f_{\theta}(x)}{T} \right) \right) \tag{13}$$

where T is the distillation temperature, which is used to control the softening degree of the teacher model output.

$$\mathcal{L}_{CE} = \frac{1}{|\mathcal{D}_i|} \sum_{(x,y)\in\mathcal{D}_i} \ell_{CE}(g_{\phi_i}(f_{\theta}(x)), y) \tag{14}$$

Furthermore, at this stage, the framework adopts a heuristic distillation temperature adjustment strategy based on model-agnostic meta-learning. To accommodate lightweight requirements, each IoT smart device tests candidate temperatures using only 10% of the local training data, with simulated time constrained. Additionally, the system dynamically narrows the candidate temperature range to $[T_{best} \pm 0.5]$ based on historical optimal values, thereby avoiding redundant testing.

- **Building a pool of temperature candidates.** Adjust candidate values according to task characteristics and use finer-grained sampling as much as possible.
- **Internal circulation simulation update.** For each $T_i \in T_{\text{candidates}}$, perform knowledge distillation simulation.
- **Select the optimal temperature for external circulation.**
- **Distilling real knowledge using the optimal temperature.**
- **Dynamic update of candidate pool.** Based on historical performance, retain high-performing T values and eliminate low-performing values to improve efficiency.

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V. EXPERIMENT

A. Experiment Setup

This paper primarily uses the MNIST [21] handwritten digit dataset, the FashionMNIST [22] fashion image dataset, and the CIFAR-10 color image dataset. The MNIST and FashionMNIST datasets are mainly used to evaluate the performance of the FedDHKD framework compared to other baseline frameworks in simple image classification tasks, while the CIFAR-10 dataset is used to test the overall performance of the FedDHKD framework in handling more complex image classification tasks.The selected dataset primarily simulates visual perception tasks in 5G IoT. The client feature extractor within the FedDHKD framework supports modification and replacement, enabling its extension to non-image data without altering the core framework. This lays the groundwork for future multimodal validation.

Considering the characteristics of strong data heterogeneity and uneven distribution in IoT, this paper sets $N = 10$ IoT smart devices as clients for federated learning. Based on a multi-factor random mechanism, the initial number of labels, label types, and sample sizes for each IoT smart device are sampled and allocated to serve as their local private training data. Due to data volume constraints, the size of the local initial dataset allocated to each IoT smart device in the experiment ranges from [2000, 10000]. In this paper, the overall data heterogeneity is fixed at weak heterogeneity and strong heterogeneity by limiting the number of labels, label types, and sample sizes allocated to each IoT smart device during dataset allocation, enabling comparison.

In terms of parameter configuration, the low-level model deployed on resource-constrained IoT smart devices is responsible for extracting high-dimensional feature representations from raw input data. It is implemented using the lightweight MobileNetV2 architecture, with an output feature dimension of 128. It ensures high feature extraction capability while maintaining low computational overhead. The high-level classification model deployed on IoT servers is responsible for classifying the noisy features uploaded by IoT devices. It uses a three-layer multi-layer perceptron (MLP) with ReLU activation functions and batch normalization, and employs a dropout layer to prevent overfitting. The input dimension is set to 128 to match the output of the low-level model, and the output dimension is 10. The number of experimental runs is set to 100, the distillation loss weight α for dynamic distillation is set to 0.7, the initial distillation temperature candidate pool is set to [1.0, 2.0, 3.0, 4.0], and the differential privacy protection technology is implemented by adding Gaussian noise. The privacy budget ϵ is set to 3, and the noise standard deviation σ is dynamically adjusted according to the dataset size and privacy requirements.

To demonstrate the performance of the FedDHKD framework, we use the following four mainstream federated learning methods as baseline comparisons: FedAvg, FedProx, SCAFFOLD, and FedMD, and design comparative experiments from multiple dimensions. Among them, FedAvg is a federated averaging method where clients upload local model parameters, and the server updates the global model using weighted averaging; FedProx introduces a proximal term constraint on top of FedAvg to mitigate model drift issues caused by heterogeneous data; SCAFFOLD incorporates a control variable mechanism to correct the local training direction of clients, enhancing model convergence under Non-IID conditions; FedMD is a federated learning method based on knowledge distillation, allowing clients to use different model structures and collaborate on training through output distributions.To ensure fairness, all comparison schemes in the experiment used structurally consistent MLP models for local training and parameter uploading, and maintained consistency in other relevant hyperparameters.

B. Experimental Result

In this subsection, we describe the experimental results used to evaluate the performance of the FedDHKD framework, aiming to validate the effectiveness of our framework in terms of model accuracy, robustness to data heterogeneity, and adaptability in resource-constrained IoT environments.

Accurary: As shown in Figure 3 and Table I, a comprehensive comparison of model performance between FedDHKD and five baselines is presented. The bar chart and line chart respectively display the test accuracy achieved by each method on the MNIST, FashionMNIST, and CIFAR-10 datasets under weak and strong heterogeneity settings, along with the accuracy improvement rate of FedDHKD relative to each baseline method. Experimental results reveal that under strong heterogeneity conditions, FedDHKD outperforms the best baseline method FedMD by 0.26%, 0.42%, and 0.38% on the MNIST, FashionMNIST, and CIFAR-10 datasets, respectively, demonstrating exceptional performance. Furthermore, as heterogeneity increases, FedDHKD exhibits a smaller accuracy decline compared to other baseline approaches, indicating stronger robustness and generalization capabilities in highly heterogeneous environments.

Communication overhead: Figure 4 shows a bar chart comparing the communication overhead of baselines in each round on different datasets. The experimental results show that FedDHKD significantly outperforms all comparison methods in terms of communication efficiency. On the MNIST and FashionMNIST datasets, FedDHKD achieves a communication overhead of only 0.3 MB per round, which is approximately 1/7 of the traditional FedAvg method, and on CIFAR-10, it achieves 0.4 MB, which is approximately 1/6 of FedAvg. FedDHKD demonstrates significant advantages in communication efficiency. Through a comprehensive design combining model partitioning, lightweight feature extraction, knowledge distillation, and differential privacy feature noise addition, FedDHKD significantly reduces communication overhead while maintaining model performance.

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TABLE I
MODEL ACCURACY (%) OF DIFFERENT METHODS UNDER MNIST, FASHIONMNIST AND CIFAR-10 DATASETS AND HETEROGENEITY LEVELS.

Method	Dataset					
	MNIST [21]		FashionMNIST [22]		CIFAR-10 [23]	
	Heterogeneity Level (Weak or Strong)					
	Weak	Strong	Weak	Strong	Weak	Strong
FedAvg [24]	96.59	93.27	88.09	84.47	64.86	59.30
DP-FedAvg [25]	95.86	92.65	87.63	84.28	64.34	58.95
FedProx [26]	96.93	94.05	88.41	85.12	65.27	59.69
SCAFFOLD [27]	97.33	94.52	88.82	85.70	65.83	60.07
FedMD [28]	97.77	95.01	89.23	86.19	66.31	60.34
FedDHKD	**97.88**	**95.26**	**89.43**	**86.55**	**66.42**	**60.57**

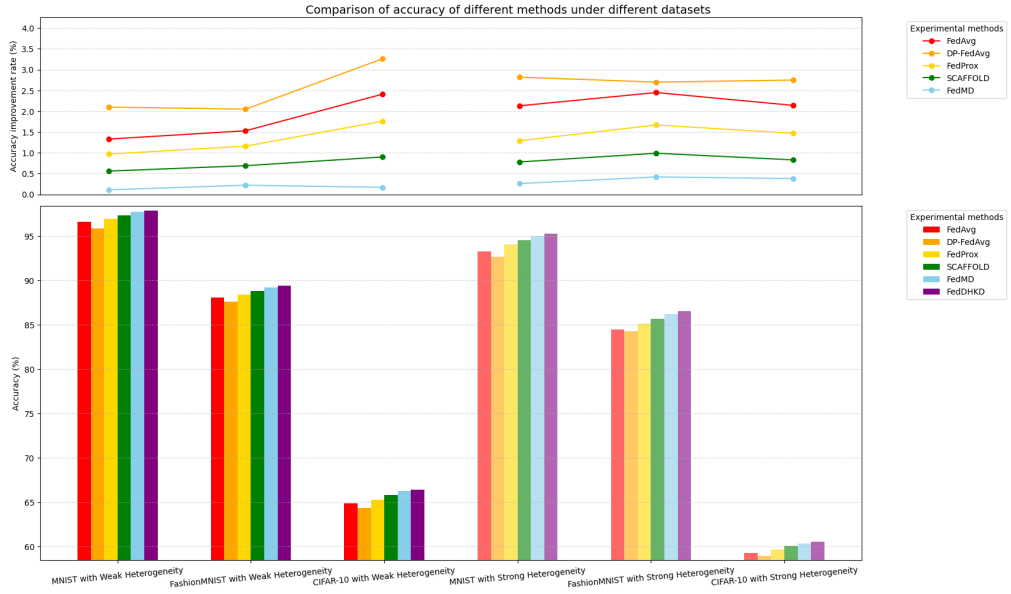


Fig. 3. Model accuracy of each scheme and improvement rate of model accuracy of the FedDHKD framework compared to baselines under MNIST, FashionMNIST and CIFAR-10 datasets.

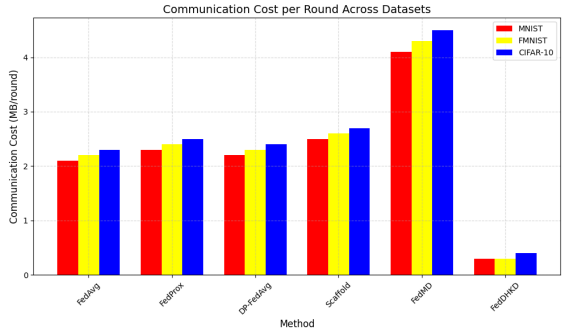


Fig. 4. Communication overhead per round for different methods under MNIST, FashionMNIST and CIFAR-10 datasets.

In contrast, methods such as FedAvg, FedProx, and Scaffold require uploading complete model parameters, while FedMD also requires uploading teacher model parameters, resulting in high communication costs. Although DP-FedAvg incorporates a differential privacy mechanism, parameter uploads still impose significant communication pressure.

Although FedDHKD achieves only 0.26% higher accuracy than FedMD under strong non-independent and identically distributed conditions, its communication overhead is reduced to one-fifth that of FedMD. This trade-off—a slight accuracy loss in exchange for substantial resource savings—precisely addresses the core requirements of 5G IoT edge devices, demonstrating significant practical value.

VI. CONCLUSION

This paper proposes a lightweight federated learning privacy protection framework based on dynamic hierarchical knowledge distillation. The framework adopts a hierarchical model architecture design, deploying the feature extraction layer on the client side and the classification decision layer

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on the server side. It achieves model compression and knowledge transfer through a dynamic knowledge distillation mechanism, effectively reducing communication costs. Additionally, it introduces an optimization strategy based on the idea of model meta-learning, enhancing the model's adaptability to the client-side data distribution. This not only improves training efficiency but also significantly alleviates the model drift issues caused by Non-IID data distributions. In terms of privacy protection, it combines feature-level differential privacy technology, adding Gaussian noise that satisfies differential privacy constraints to the low-dimensional features that need to be uploaded, providing lightweight and effective data privacy protection. This framework combines model partitioning, dynamic hierarchical knowledge distillation, model-agnostic meta-learning-inspired heuristic optimization strategies, and differential privacy protection techniques. It reduces computational and communication overhead on edge devices while maintaining model performance, offering lightweight privacy protection capabilities. This provides an efficient solution to address the challenges faced by traditional federated learning in 5G IoT scenarios, such as resource-constrained edge devices, strong data distribution heterogeneity, and high privacy protection requirements, demonstrating significant research value and application potential.

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